

Danyal Mohaddes

Lead research scientist · Computational physics · Complex reacting flows

Boston, MA · [danyal@mohaddes.dev] · [Google Scholar] · [LinkedIn]

Mechanical engineering Ph.D. developing first-principles, high-fidelity simulations as well as reduced-order models of complex, multiphase, highly-coupled thermo-chemical and fluid-dynamical phenomena. Experienced in implementing and deploying efficient code on HPC architectures to generate large datasets, and validating results against test data for high-temperature, safety-critical hardware and systems.

EDUCATION

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| Stanford University | Stanford, USA |
| Ph.D. in Mechanical Engineering | 2017–2022 |
| <ul style="list-style-type: none">– Thesis: <i>Numerical Simulation of Hot Surface Ignition and Combustion of Fuel Sprays</i>– Advisor: Prof. Matthias Ihme– Coursework: Numerical mathematics, reacting flow physics, parallel and GPU programming, deep learning | |
| Stanford University | Stanford, USA |
| M.Sc. in Mechanical Engineering | 2016–2018 |
| <ul style="list-style-type: none">– Concentration: Flow Physics and Computational Engineering | |
| University of Toronto | Toronto, Canada |
| B.A.Sc. in Mechanical Engineering (High Honors, 3.9/4.0) | 2011–2016 |
| <ul style="list-style-type: none">– Concentrations: Energy, Solid mechanics | |

EXPERIENCE

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| FM, Research Division | Boston, USA |
| Lead Research Scientist, Fire Modeling | 2022– |
| <i>Promoted from Senior Research Scientist in 2026</i> | |
| <ul style="list-style-type: none">– Developed and implemented (OpenFOAM/C++) physics-based computational models of fire propagation, conjugate heat transfer, solid fuel pyrolysis and lithium-ion battery (LIB) thermal runaway for deployment on cloud-based (AWS) and on-premises high-performance computing (HPC) architectures [Open Source]– Derived the first theoretical model of thermal runaway propagation (TRP) in LIB stacks [Paper]; collaborated with experimentalists on design and analysis of validation experiments, as well as numerical optimization for parameter estimation from experimental data [Paper] [Open Source]– Developed a large language model (LLM) agent to inspect simulation software using retrieval-augmented generation (RAG), and to set up, execute, and post-process fire dynamics simulations on HPC systems. Led and mentored an intern who contributed to the implementation. [Paper]– Conducted multi-GPU training and development of a machine learning (ML) surrogate model in PyTorch for radiation transport leveraging deep operator networks (DeepONets) and CFD datasets– Developed generative (GAN) and regressive (CNN) ML approaches for physical modeling of multiphase reacting fluid flows and applied foundation models for image segmentation for experimental data analysis– Served as the representative for the Research Division on FM's Responsible AI (RAI) Oversight Group, which gives guidance and feedback to teams implementing major AI projects across the company– Implemented a development workflow across multiple research teams in Azure DevOps: version control, issue tracking, automated regression testing pipeline, containerization (Docker), continuous integration, documentation | |

- Arbury Labs, Inc. Seattle, USA
Scientific Advisor (Remote) 2023–2024
- Provided research services to a startup developing a coaching application for sports leveraging computer vision and LLMs
 - Developed interfaces to LLMs for app integration and implemented one/few-shot learning approaches for data analysis, human-machine-human interaction and feedback, synthetic data generation and program logic
- Stanford University Stanford, USA
Graduate Research Assistant under Prof. Matthias Ihme 2016–2022
- Extended a dynamic modeling framework and applied a global optimization algorithm for computationally efficient and physically accurate modeling of multiphase chemical effects in combustion simulations [Paper]
 - Conducted high-fidelity modeling of flow physics using massively-parallel simulations; validated simulation results through non-dimensional scaling and analysis of experimental data [Paper], [Paper]
 - Performed parallelized (C++/MPI/OpenMP) simulations of a Lagrangian-Eulerian system with over 6M degrees of freedom to model ignition of fuel leaks in aircraft in collaboration with Boeing [Paper]
 - Developed a parallelized (C++/OpenMP) solver for highly stiff and coupled combustion PDEs [Open Source]
 - Employed ML-based data analysis techniques in Python (TensorFlow) to analyze parametric sensitivities in numerical simulations of combustion problems [Paper]
 - Developed thermodynamic and heat transfer models of porous media combustion for improved gas turbine engine performance in aerospace applications with NASA Glenn [Paper]
 - Designed, machined, instrumented, and constructed custom test stands with LabVIEW interfaces for porous media and laminar flow combustion experiments [Paper], [Paper], [Paper]
- Technische Universität München Munich, Germany
Visiting Student Researcher under Prof. Oskar Haidn Summer 2019
- Performed high-fidelity numerical modeling for gas turbine and rocket propulsion using C++-based software [Paper]
 - Supported by Stanford Graduate Internship and Research Program in Germany
- Amec Foster Wheeler (now Wood plc.) Toronto, Canada
Engineering Intern, Canadian Nuclear Division Summer 2015
- Programmed databases in Visual Basic for Applications (VBA) for quantified risk analysis of fluid systems in thermal and nuclear power plants
- Syncrude Canada Ltd. Edmonton & Fort McMurray, Canada
Research Intern, Division of Research and Development 2014-2015
- Performed structural finite-element analysis of process and mechanical components using ANSYS Workbench
 - Performed economic analysis and developed business cases for upgrades to heavy industrial machinery
 - One week per month, served as an assistant technician in the oil fields, installing sensors and performing maintenance of monitoring equipment on live turbomachinery and process piping
- University of Toronto Toronto, Canada
Undergraduate Research Assistant under Prof. David Sinton and Prof. Nasser Ashgriz Summer 2013/2014
- Developed image-processing software for identifying and tracking droplet behavior in nuclear engineering experiments [Open Source, over 700 downloads]
 - Developed and conducted microfluidic experiments quantifying the minimum miscibility pressure of petroleum with CO₂, for CO₂ enhanced oil recovery and sequestration [Paper]

- Developed software and performed 3D graphics design for a custom C++-based virtual reality simulator for experiments in the early detection of Alzheimer’s disease [Paper]

AWARDS, GRANTS AND HONORS

Most Collaborative Participant, Stanford FLAME AI Workshop	2023
Boeing Research & Technology Strategic Universities Program, <i>grant awarded to PI’s lab to support my PhD</i>	2018–2022
Natural Sciences and Engineering Research Council of Canada (NSERC) Postgraduate Scholarship	2019–2020
Graduate Internship and Research Program in Germany	2019
Departmental Graduate Engineering Fellowship Award	2016–2017
Editors’ Choice article published in <i>Analytical Chemistry</i>	2014
Eight undergraduate research and academic scholarships	2011–2016
<i>NSERC summer research award, U. Toronto Excellence Award for summer research, ASME student scholarship, SPE Canada student scholarship, SPE Calgary section student scholarship, Otto Holden scholarship, WJT Wright scholarship, Faculty admission scholarship</i>	

PUBLICATIONS

Journal papers

- Ren, N., Lu, X., **Mohaddes, D.**, Krisman, A., Wang, Y., “A dynamic method for overcoming the ray effects of dom radiation model in fire spread simulations,” *Proceedings of the Combustion Institute*, vol. 42, p. 106 339, 2026.
- Wang, Y., Xiong, G., Zeng, D., **Mohaddes, D.**, “Towards fire hazard evaluation of stationary lithium-ion battery systems: A review,” *Fire Safety Journal*, p. 104 856, 2026. DOI: 10.1016/j.firesaf.2026.104856
- Mohaddes, D.**, Wang, Y., “Theory and analysis of module-scale thermal runaway propagation,” *Combustion and Flame*, vol. 279, p. 114 327, 2025. DOI: 10.1016/j.combustflame.2025.114327
- Zeng, D., **Mohaddes, D.**, Gagnon, L., Wang, Y., “Modeling initiation and propagation of thermal runaway in pouch li-ion battery cells: Effects of heating rate and state-of-charge,” *Proceedings of the Combustion Institute*, vol. 40, no. 1, p. 105 316, 2024. DOI: 10.1016/j.proci.2024.105316
- Mohaddes, D.**, Brouzet, D., Ihme, M., “Cost-constrained adaptive simulations of transient spray combustion in a gas turbine combustor,” *Combustion and Flame*, vol. 249, p. 112 530, 2023. DOI: 10.1016/j.combustflame.2022.112530
- Mohaddes, D.**, Ihme, M., “Wall heat transfer and flame structure transitions in stagnating spray flames,” *Proceedings of the Combustion Institute*, vol. 39, no. 2, pp. 2683–2692, 2023. DOI: 10.1016/j.proci.2022.08.037
- Mohaddes, D.**, Ihme, M., “On the hot surface ignition of a wall-stagnating spray flame,” *Combustion and Flame*, vol. 240, p. 111 988, 2022. DOI: 10.1016/j.combustflame.2022.111988
- Dodd, M. S., **Mohaddes, D.**, Ferrante, A., Ihme, M., “Analysis of droplet evaporation in isotropic turbulence through droplet-resolved DNS,” *International Journal of Heat and Mass Transfer*, vol. 172, p. 121 157, 2021. DOI: 10.1016/j.ijheatmasstransfer.2021.121157
- Mohaddes, D.**, Boettcher, P., Ihme, M., “Hot surface ignition of a wall-impinging fuel spray: Modeling and analysis using large-eddy simulation,” *Combustion and Flame*, vol. 228, pp. 443–456, 2021. DOI: 10.1016/j.combustflame.2021.02.025

Mohaddes, D., Xie, W., Ihme, M., “Analysis of low-temperature chemistry in a turbulent swirling spray flame near lean blow-out,” *Proceedings of the Combustion Institute*, vol. 38, no. 2, pp. 3435–3443, 2021. DOI: 10.1016/j.proci.2020.08.030

Sobhani, S., Muhunthan, P., Boigné, E., **Mohaddes, D.**, Ihme, M., “Experimental feasibility of tailored porous media burners enabled via additive manufacturing,” *Proceedings of the Combustion Institute*, vol. 38, no. 4, pp. 6713–6722, 2021. DOI: 10.1016/j.proci.2020.06.120

Watson, L. M., Dunham, E. M., **Mohaddes, D.**, Labahn, J., Jaravel, T., Ihme, M., “Infrasound Radiation From Impulsive Volcanic Eruptions: Nonlinear Aeroacoustic 2D Simulations,” *Journal of Geophysical Research: Solid Earth*, vol. 126, no. 9, pp. 1–28, 2021. DOI: 10.1029/2021JB021940

Mohaddes, D., Chang, C. T., Ihme, M., “Thermodynamic cycle analysis of superadiabatic matrix-stabilized combustion for gas turbine engines,” *Energy*, vol. 207, p. 118 171, 2020. DOI: 10.1016/j.energy.2020.118171

Boigne, E., Muhunthan, P., **Mohaddes, D.**, Wang, Q., Sobhani, S., Hinshaw, W., Ihme, M., “X-ray computed tomography for flame-structure analysis of laminar premixed flames,” *Combustion and Flame*, vol. 200, pp. 142–154, 2019. DOI: 10.1016/j.combustflame.2018.11.015

Sobhani, S., **Mohaddes, D.**, Boigne, E., Muhunthan, P., Ihme, M., “Modulation of heat transfer for extended flame stabilization in porous media burners via topology gradation,” *Proceedings of the Combustion Institute*, vol. 37, no. 4, pp. 5697–5704, 2019. DOI: 10.1016/j.proci.2018.05.155

Nguyen, P., **Mohaddes, D.**, Riordon, J., Fadaei, H., Lele, P., Sinton, D., “Fast Fluorescence-Based Microfluidic Method for Measuring Minimum Miscibility Pressure of CO₂ in Crude Oils,” *Analytical Chemistry*, vol. 87, no. 6, pp. 3160–3164, 2015. DOI: 10.1021/ac5047856

Byagowi, A., **Mohaddes, D.**, Moussavi, Z., “Design and Application of a Novel Virtual Reality Navigational Technology (VRNChair),” *Journal of Experimental Neuroscience*, vol. 8, no. 1, JEN.S13448, 2014. DOI: 10.4137/JEN.S13448

PhD thesis

Mohaddes, D., *Numerical simulation of hot surface ignition and combustion of fuel sprays*, Stanford University, 2022.

Conference podium presentations

Mohaddes, D., *CFD modeling of LIB thermal runaway in a BESS rack*, FM Fire Modeling Workshop, 2026.

Mohaddes, D., Wang, Y., *Fire dynamics and inter-module thermal runaway propagation in battery energy storage systems*, International Conference on Numerical Combustion, 2025.

Mohaddes, D., Wang, Y., *Module-scale thermal runaway propagation: Theory and analysis*, 14th US National Combustion Meeting, 2025.

Mohaddes, D., Wang, Y., *Toward fire dynamics simulations of thermal runaway propagation in battery energy storage systems*, 20th OpenFOAM Workshop, 2025.

Mohaddes, D., Wang, Y., *Toward simulation of thermal runaway in battery energy storage systems using firefoam*, FM Fire Modeling Workshop, 2025.

Mohaddes, D., Zeng, D., Gagnon, L., Krisman, A., Ren, N., *Lithium-ion battery modeling: From cell-level thermal runaway to multi-module fires*, FM Global Open Source CFD Fire Modeling Workshop, 2024.

Mohaddes, D., *Numerical simulation of hot surface ignition and combustion of fuel sprays*, Stanford University, 2022.

Mohaddes, D., Ihme, M., *Adaptive combustion modeling of a gas turbine engine near lean blow-out*, International Conference on Numerical Combustion, 2022.

- Mohaddes, D.**, Ihme, M., *Analysis of the hot surface ignition limits of a wall-stagnating fuel spray*, Spring Meeting of the Western States Section of the Combustion Institute, 2022.
- Mohaddes, D.**, Ihme, M., *Wall heat transfer and flame structure transitions in stagnating spray flames*, 39th International Symposium on Combustion, 2022.
- Mohaddes, D.**, Ihme, M., *Parametric study of hot surface ignition of an impinging fuel spray using large-eddy simulation*, APS Division of Fluid Dynamics Annual Meeting, 2020.
- Mohaddes, D.**, Xie, W., Ihme, M., *Analysis of low-temperature chemistry in a turbulent swirling spray flame near lean blow-out*, 38th International Symposium on Combustion, 2020.
- Mohaddes, D.**, Ihme, M., *Resolution requirements for LES modeling of a methanol pool fire*, FM Global Open Source CFD Fire Modeling Workshop, 2019.
- Mohaddes, D.**, Ihme, M., Damazo, J., Boettcher, P., Moravec, B., *Computational modeling of compartment fires for aircraft safety*, International Conference on Numerical Combustion, 2019.
- Mohaddes, D.**, Xie, W., Ihme, M., *Flame structure analysis and flame stabilization in a turbulent swirling spray flame*, APS Division of Fluid Dynamics Annual Meeting, 2019.
- Mohaddes, D.**, Ihme, M., *Computational modeling of accidental fire spread in under-ventilated compartments*, APS Division of Fluid Dynamics Annual Meeting, 2018.
- Mohaddes, D.**, Sobhani, S., Boigne, E., Muhunthan, P., Ihme, M., *Experimental investigation of flame stability in porous media burners*, APS Division of Fluid Dynamics Annual Meeting, 2017.
- Mohaddes, D.**, Asghriz, N., *Study of vapor bubbles in horizontal PHWR cores*, Canadian Society for Mechanical Engineering International Congress, 2014.

Other conference contributions

- Xiao, T., **Mohaddes, D.**, Brown, W., Xiong, G., Wang, Y., *Modeling of small pouch li-ion battery cells: Thermal runaway initiation and propagation*, 14th US National Combustion Meeting, 2025.
- Xu, L., **Mohaddes, D.**, Wang, Y., “LLM agent for fire dynamics simulations,” in *Neurips 2024 Workshop Foundation Models for Science: Progress, Opportunities, and Challenges*, 2024.
- Zeng, D., **Mohaddes, D.**, Gagnon, L., Wang, Y., *Modeling initiation and propagation of thermal runaway in pouch Li-ion battery cells: Effects of heating rate and state-of-charge*, 40th International Symposium on Combustion, 2024.
- Mohaddes, D.**, Lu, X., Krisman, A., Ren, N., *Devops for fire modeling research*, FM Global Open Source CFD Fire Modeling Workshop, Poster, 2023.
- Mohaddes, D.**, Ren, N., Wang, Y., *Firefoam simulation of the macfp-3 parallel-panel target*, 3rd Workshop of the Measurement and Computation of Fire Phenomena (MaCFP) Database, Poster, 2023.
- Bonanni, M., **Mohaddes, D.**, Ly, N., Perakis, N., Hardi, J., Börner, M., Ihme, M., “Toward Numerical Investigation of Ignition and Combustion Transition in a Subscale LOX/Methane Rocket Combustor,” in *AIAA Scitech Forum*, 2021. DOI: 10.2514/6.2021-1141
- Sobhani, S., Muhunthan, P., Boigné, E., **Mohaddes, D.**, Ihme, M., *Experimental feasibility of tailored porous media burners enabled via additive manufacturing*, 38th International Symposium on Combustion, 2020.
- Muhunthan, P., Sobhani, S., Boigne, E., **Mohaddes, D.**, Ihme, M., *Experimental investigation of combustion in porous media burners with tailored matrix-structure using additive manufacturing*, APS Division of Fluid Dynamics Annual Meeting, 2019.
- Sobhani, S., Muhunthan, P., **Mohaddes, D.**, Boigne, E., Cheng, Z., Ihme, M., *Enabling tailored porous media burners via additive manufacturing*, Proceedings of the 11th US National Combustion Meeting, 2019.

Ihme, M., **Mohaddes, D.**, *Toward the modeling and analysis of oxygen-controlled turbofan engine fan case compartment fires*, FM Global Open Source CFD Fire Modeling Workshop, 2018.

Sobhani, S., **Mohaddes, D.**, Boigne, E., Muhunthan, P., Ihme, M., *Modulation of heat transfer for extended flame stabilization in porous media burners via topology gradation*, 37th International Symposium on Combustion, 2018.

Boigne, E., Muhunthan, P., **Mohaddes, D.**, Sobhani, S., Parkinson, D., Barnard, H., Ihme, M., *Pore-scale and topology analysis of flame stabilization inside inert porous media using x-ray microtomography*, APS Division of Fluid Dynamics Annual Meeting, 2017.

Muhunthan, P., Sobhani, S., Boigne, E., **Mohaddes, D.**, Hinshaw, W., Ihme, M., *Calibration of x-ray computed tomography (xct) using a flat flame burner*, APS Division of Fluid Dynamics Annual Meeting, 2017.

Sobhani, S., Muhunthan, P., Boigne, E., **Mohaddes, D.**, Ihme, M., *Investigation of pore-scale flow physics in porous media burners*, APS Division of Fluid Dynamics Annual Meeting, 2017.

Byagowi, A., **Mohaddes, D.**, McLeod, R. D., *Accidental emergence within an agent based model: Simulation of agent interactions in an emergency situation*, Proceedings of the 17th International Conference on Computer Games (CGAMES), 2012. DOI: 10.1109/CGames.2012.6314574

Public repositories (see github.com/danmohad for complete list)

- LLMAgora* – (Python, 2026) A flexible software arena in which LLM-agents can interact and debate.
- saving-vs-inheritance* – (Python/TeX, 2026) Software and text source for the working paper “Can saving catch inheritance? A normalized map of wealth catch-up”.
- signed-input-capture* – (C/Python, 2026) Software and hardware project for cryptographic input buffer verification.
- khod-kaar* – (Python, 2023) An early implementation of an LLM-based SWE agent with shell control from scratch.
- spray-HSI* – (C++, 2021) Parallelized solver for unsteady multiphase reacting flows that are wall-impinging.

PROFESSIONAL SERVICE

Peer review

Proceedings of the Combustion Institute	9 reviews since 2022
Fire Safety Journal	7 reviews since 2023
Combustion and Flame	4 reviews since 2021
Combustion Science and Technology	2 reviews since 2023
International Journal of Thermal Sciences	1 review since 2024
International Journal of Hydrogen Energy	1 review since 2023
Fuel	1 review since 2022
Journal of Engineering for Gas Turbines and Power	2 reviews since 2022
AIMS Energy	1 review since 2021

Session chair appointments

14th US National Combustion Meeting: Computational and ML/AI-Data Driven Combustion	2025
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Mentorship

Leidong Xu, AI/ML Ph.D. summer intern at FM Research	2024
Matthew Bonanni, Ph.D. candidate at Stanford	2019–2022
Wai Tong Chung, Ph.D. candidate at Stanford	2018–2022

Advisorship

Stanford Graduate School of Business, Gukomeza Sprayer team, Technical advisor

2018

TEACHING AND INVITED TALKS

<i>Advances in theory and modeling of thermal runaway in lithium-ion battery systems across scales</i> , U. Delaware Battery Safety Workshop (Invited talk)	2026
<i>From Combustion Fundamentals to Fire Modeling</i> , Combustion Processes, Cornell University (Guest Lecture)	2024
<i>Computational Fluid Dynamics in C++</i> , Computing for Geomatics Engineers, U. Calgary (Guest Lecture)	2023
<i>Fidelity-Adaptive Modeling for Spray Combustion</i> , Saudi Aramco (Invited talk)	2022
<i>Modeling Hot Surface Ignition with C++</i> , Computing for Geomatics Engineers, U. Calgary (Guest Lecture)	2022
<i>Modern Applications of Combustion Science</i> , Combustion Processes, Cornell University (Guest Lecture)	2021
<i>Hot Surface Ignition of Wall-Impinging Fuel Sprays</i> , FM Global Research (Job Talk)	2021
Gas Turbine Design and Analysis, Stanford University (Teaching Assistant)	2020

SKILLS

Technology

Programming: Object-oriented C/C++, Python, Matlab

HPC: MPI, OpenMP, CUDA

Frameworks: Numpy, SciPy, Pandas, scikit-learn, TensorFlow, PyTorch, Cantera

Toolbox: Linux, vim, git, bash, zsh, slurm, mongoddb, Docker

Design: SolidWorks, ANSYS

Experiments: LabVIEW, Arduino, Raspberry Pi, BeagleBone

AI-driven coding: Claude Code, Codex

Human Languages

English: Native language

French: Fluent (C1/C2 level)

German: Fluent (C1 level)

Persian: Fluent (native-level speaking ability, competent in reading and writing)

INTERESTS AND ACTIVITIES

Linguistics, especially phonology and writing systems; poetry and its analysis, especially ancient; history, foreign policy analysis and grand strategy; freestyle wrestling; classical guitar.